

Script-Controlled ACL - Restrict Record Access Based on Field Value

ServiceNow Micro-Project

PHASE 6 | PROJECT TESTING PHASE

6. Project Testing Phase

Testing was performed by using ServiceNow's **Impersonate user** feature to switch between the EEE User (with an incrementally increasing set of roles) and the Administrator, then observing the Institution Details list at `u_institution_details_list.do` for each scenario.

6.1 Test Case Summary

Test ID	Scenario	Roles Held by EEE User	Expected Result	Actual Result	Status
TC-01	Read access with branch role	bb1	Only EEE-branch records visible	Only EEE-branch records visible	PASS
TC-02	Read access without any role	none	No records displayed	"No records to display" shown	PASS
TC-03	Admin read access	admin	All records (ECE, EEE, CSE) visible	All records visible	PASS
TC-04	Create access	bb1, bb2	EEE records visible + New button available	EEE records visible with New button	PASS
TC-05	Write (edit) access	bb1, bb2, bb3	EEE records editable in place (e.g., Description field)	Description field editable and saved	PASS
TC-06	Delete access	bb1, bb2, bb3, bb4	Selected EEE record can be deleted via Actions menu	Record deleted after confirmation	PASS

6.2 TC-01 & TC-02 — Read ACL Verification

Steps: Impersonate EEE User (role: bb1) → open Institution Details list.

institution Details

Student Roll Numbe

Search

Actions on selected rows...

All

☐	Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
	INS0001007	EEE		saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
	INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

Output screen — User with the bb1 role can view only the two EEE-branch records (INS0001007, INS0001004).

Steps: Remove bb1 (or impersonate a user without any bb role) → open Institution Details list.

institution Details

Student Roll Numbe

Search

Search

Student Roll Number

Branch

Description

Email

Faculty name

Phone Number

Student Name

Search

Search

Search

Search

Search

Search

Search

?

No records to display

Output screen — A user without the required role cannot view any records ("No records to display").

6.3 TC-03 — Administrator Override Verification

Steps: Impersonate Administrator → open Institution Details list.

institution Details

Student Roll Numbe

Search

Actions on selected rows...

New

All

Student Roll Number	Faculty name	Student Name	Branch	Email	Phone Number
Search	Search	Search	Search	Search	Search
INS0001006	ITIL User	Praneeth	ECE	praneeth@gmail.com	8989090906
INS0001008	Malla Sravan kumar	Abraham Lincoln	CSE	sravan@gmail.com	8978908989
INS0001005	ITIL User	Srinu Nandipalli	ECE	srinu@gmail.com	9089909090
INS0001007	Abel Tuter	Guna Sai	EEE	saiguna@gmail.com	9090898765
INS0001001	Abraham Lincoln	Yaswanth Nikku	CSE	yaswanth@gmail.com	8989898989
INS0001004	Abel Tuter	Prashanth Reddy	EEE	prashanth@gmail.com	9090909090

Output screen — Administrator can view all records regardless of branch (ECE, CSE, EEE all shown).

6.4 TC-04 — Create ACL Verification

Steps: Assign bb2 in addition to bb1 → reopen the list.

institution Details

Student Roll Numbe

Search

Actions on selected rows...

New

All

Student Roll Number

Branch

Description

Email

Faculty name

Phone Number

Student Name

INS0001007

EEE

saiguna@gmail.com

Abel Tuter

9090898765

Guna Sai

INS0001004

EEE

prashanth@gmail.com

Abel Tuter

9090909090

Prashanth Reddy

Output screen — User with bb1 & bb2 roles can view EEE-branch records and now has the New button available for record creation.

6.5 TC-05 — Write ACL Verification

Steps: Assign bb3 in addition to bb1/bb2 → edit the Description field of an EEE record inline.

institution Details

Student Roll Numbe

Search

Actions on selected rows...

New

All

☐	Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
	INS0001007	EEE		com	Abel Tuter	9090898765	Guna Sai
	INS0001004	EEE		il.com	Abel Tuter	9090909090	Prashanth Reddy

Output screen — Description field becomes editable for the user holding bb1, bb2 and bb3 roles.

institution Details

Student Roll Numbe

Search

Actions on selected rows...

New

All

☐	Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
☐	INS0001007	EEE	ACL is Executed	saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
☐	INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

Output screen — Edit successfully saved ("ACL is Executed") confirming Write ACL is functioning correctly.

6.6 TC-06 — Delete ACL Verification

Steps: Assign bb4 in addition to bb1/bb2/bb3 → select a record → Actions on selected rows → Delete → confirm.

institution Details

Student Roll Numbe

Search

Actions on selected rows...

New

All

Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
INS0001007	EEE	ACL is Executed	saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

Output screen — Record selected for deletion by the user holding all four roles (bb1-bb4).

institution Details

Student Roll Numbe

Search

Actions on selected rows...

New

All

Student Roll Number

Branch

Description

Email

Faculty name

Phone Number

Student Name

INS0001007

EEE

ACL is Executed

saiguna@gmail.com

Abel Tuter

90908987

Guna Sai

INS0001004

EEE

ACL is Executed

prashanth@gmail.com

Abel Tuter

90909090

Prashanth Reddy

Actions on selected rows...

Delete

Delete with preview...

Create Application File

Assign Tag:

Now tag

Now Platform Security

Strategic Portfolio Management

Field Service Management

Now Platform

Security

IT Asset Management

IT Service Management

IT Operations Management

More...

Output screen — Delete action confirmed, completing full CRUD verification for the EEE-branch record.

6.7 Regression Observations

- No non-EEE record was ever exposed to the EEE User at any role stage, confirming the Data Condition (Branch is EEE) correctly scopes the Read ACL.
- Removing a role immediately removed the corresponding capability on next list refresh, confirming ACL evaluation is real-time and not cached against stale permissions.
- The Administrator was never blocked at any stage, confirming the `gs.hasRole('admin')` check in each script functions as the intended override.

All six test cases passed. The script-controlled ACL implementation correctly restricts read, create, write, and delete operations on the Institution Details table to users holding the appropriate branch-specific role, while preserving full administrator access.